Temporal Counterfactual Credit Assignment in Cooperative Multi-Agent Reinforcement Learning

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***Abstract*—In cooperative multi-agent reinforcement learning (MARL), effectively distributing credit among agents remains a fundamental challenge. While the Counterfactual Multi-Agent (COMA) framework addresses spatial credit assignment by isolating each agent's marginal contribution to a joint reward at a single timestep, it relies on a TD(0) critic target that can be unstable during early training and fails to capture the delayed temporal impact of agent actions. This paper introduces Temporal Counterfactual Credit Assignment (TCCA), a novel method that extends the COMA counterfactual baseline by substituting the TD(0) target with a k-step discounted return computed from replay buffer trajectories. This gives each agent a richer credit signal reflecting not just immediate but long-horizon consequences of its actions. We integrate TCCA with Soft Actor-Critic (SAC) to add entropy regularization and off-policy training stability, and evaluate on two cooperative benchmarks: PettingZoo MPE `simple_spread_v3` and StarCraft II SMAC `3m`. Empirically, TCCA with k=5 converges to a final reward of -21 on MPE — outperforming COMA's -23 — while avoiding COMA's catastrophic critic crash at episode 8,000. On SMAC, TCCA-SAC achieves a peak win rate of $\sim$90% versus COMA's plateau of $\sim$75%, with notably faster early convergence. Ablations confirm a monotonic benefit from increasing k, validating the temporal credit hypothesis.**

***Index Terms*—multi-agent reinforcement learning, credit assignment, counterfactual baselines, soft actor-critic, StarCraft II, cooperative agents**

I. Introduction

Cooperative multi-agent reinforcement learning (MARL) studies settings where a group of agents must learn coordinated behaviors to maximize a shared objective [7]. A central difficulty is the *credit assignment problem*: when only a global team reward R_t is observed, individual agents cannot directly determine how much their own actions contributed to the outcome. This ambiguity produces noisy learning signals and can severely impede convergence, especially in environments where any single agent's contribution is diluted by the actions of others.

Existing solutions approach this problem along two axes. *Spatial* credit assignment methods — most notably COMA [1] — disentangle each agent's marginal contribution at a given timestep by constructing a counterfactual baseline: the expected value the team would have obtained had agent i acted according to its current policy while all others acted as observed. The difference between the actual joint Q-value and this baseline gives an agent-specific advantage signal. However, COMA estimates Q-values using TD(0) Bellman targets, which are only meaningful once the critic has converged. Early in training, this leads to a well-known instability where the critic produces inaccurate targets and the actor updates in unhelpful directions.

Temporal credit assignment — understanding how actions taken now affect rewards in the distant future — is largely orthogonal to spatial methods and comparatively under-explored in the counterfactual credit assignment literature. Methods such as QMIX [2] decompose value functions but do not model counterfactual scenarios; SAC [3] operates in single-agent settings without per-agent credit decomposition.

This paper proposes **Temporal Counterfactual Credit Assignment (TCCA)**, which bridges this gap. The key idea is simple: instead of substituting a TD(0) Q-estimate into the COMA advantage, we substitute a k-step discounted return $G_t^{(k)}$ computed directly from observed rewards in a replay buffer. This return is unbiased for the first k steps (real rewards, no approximation) and falls back to a critic bootstrap only at the horizon $t{+}k$. The parameter k controls the bias-variance tradeoff: small k trusts the critic more; large k trusts observed rewards more [4].

We further integrate TCCA with Soft Actor-Critic (SAC) [3], using a twin-critic architecture, entropy regularization, and Polyak-averaged target networks to stabilize off-policy learning. The resulting TCCA-SAC system is evaluated against COMA baselines on MPE and SMAC, with ablations over the temporal horizon k and the SAC entropy coefficient α.

Our main contributions are:

- A formal definition of the TCCA advantage signal and its integration with the COMA counterfactual baseline.
- A full TCCA-SAC implementation for partially observable environments using GRU actors and twin critics.
- Empirical validation on MPE and SMAC showing consistent improvement over COMA in both final performance and sample efficiency.
- Ablation studies confirming that increasing temporal horizon k monotonically improves performance.

II. Background and Related Work

A. Counterfactual Multi-Agent Policy Gradients (COMA)

Foerster et al. [1] introduced COMA to address spatial credit assignment in cooperative MARL. A centralized critic computes the joint action-value $Q(s_t, \mathbf{a}_t)$ under the global

state s_t and joint action $\mathbf{a}_t = (a_1^t, \ldots, a_n^t)$. For agent i, the counterfactual baseline is:

$$B_i(s_t, \mathbf{a}_{-i}^t) = \sum_{a_i'} \pi_i(a_i' \mid o_i^t)\, Q\big(s_t,\, (\mathbf{a}_{-i}^t,\, a_i')\big) \tag{1}$$

where $\mathbf{a}_{-i}^t$ denotes the observed actions of all agents except i. The counterfactual advantage is then:

$$A_i^{\mathrm{CF}} = Q(s_t, \mathbf{a}_t) - B_i(s_t, \mathbf{a}_{-i}^t) \tag{2}$$

This baseline answers: what would the expected value be if agent i acted according to its current policy, while others acted as observed? Crucially, $Q(s_t, \mathbf{a}_t)$ in COMA is estimated via TD(0) — a one-step Bellman target:

$$y_t^{\mathrm{TD}(0)} = R_t + \gamma\, Q(s_{t+1}, \mathbf{a}_{t+1}) \tag{3}$$

This limits the temporal reach of the credit signal and makes the advantage estimate sensitive to early critic inaccuracy.

B. QMIX: Monotonic Value Decomposition

Rashid et al. [2] propose decomposing the team value through a monotonic mixing network:

$$Q_{\mathrm{tot}}(s, \mathbf{a}) = f_\phi(Q_1(o_1, a_1),\, \ldots,\, Q_n(o_n, a_n)) \tag{4}$$

Each agent maintains a local value function Q_i, while f_ϕ combines them subject to the IGM (Individual-Global-Max) constraint. QMIX does not construct counterfactual baselines and cannot evaluate the impact of a single agent's deviation.

C. Soft Actor-Critic (SAC)

Haarnoja et al. [3] propose SAC as an off-policy actor-critic algorithm maximizing both expected reward and policy entropy:

$$J(\pi) = \mathbb{E}_{\tau\sim\pi}\left[\sum_t \gamma^t \big(R_t + \alpha\, \mathcal{H}(\pi(\cdot \mid o_t))\big)\right] \tag{5}$$

where $\mathcal{H}(\pi)$ is the policy entropy and α controls exploration. SAC uses a replay buffer and Polyak-averaged target networks, achieving strong stability on continuous control tasks. We adapt it to discrete actions following [3].

D. Gap in Existing Literature

None of the above methods jointly address temporally-extended counterfactual credit assignment in an off-policy, entropy-regularized setting. COMA is on-policy and uses instantaneous critic estimates; QMIX lacks counterfactual baselines; SAC operates without per-agent credit decomposition. TCCA bridges these gaps.

III. Proposed Method: TCCA

A. Motivation

Consider an agent that takes a positioning action at time t that carries little immediate reward but enables a decisive coordination opportunity at time $t+3$. COMA's TD(0) advantage A_i^{CF} can only reflect this if the critic has already learned to propagate the future value backward — which, early in training, it has not. The result is that TD(0) targets are unreliable precisely when they are needed most: during the initial learning phase.

Using a multi-step return $G_t^{(k)}$ computed from observed rewards directly avoids this dependence. The first k steps contribute *actual* rewards from the environment; only the terminal bootstrap $V(s_{t+k})$ relies on the critic. As training progresses and the critic improves, the bootstrap term becomes more reliable — but by then, the return estimate is already anchored by k real reward observations.

B. Temporal Return Estimation

At training time, we sample consecutive trajectory segments of length k from a replay buffer:

$$\tau_t = \{(s_t, \mathbf{a}_t, R_t),\ (s_{t+1}, \mathbf{a}_{t+1}, R_{t+1}),\ \ldots,\ (s_{t+k}, \mathbf{a}_{t+k}, R_{t+k})\} \tag{6}$$

The k-step discounted return from step t is:

$$G_t^{(k)} = \sum_{l=0}^{k-1} \gamma^l R_{t+l} \;+\; \gamma^k V(s_{t+k}) \tag{7}$$

where the bootstrap value $V(s_{t+k})$ is the expected Q-value under the current policy:

$$V(s_{t+k}) = \sum_a \pi_i(a \mid o_{t+k})\, Q_\phi(s_{t+k},\, a,\, \mathbf{a}_{-i,t+k}) \tag{8}$$

In the SAC variant, the soft bootstrap replaces the standard value with the entropy-regularized form:

$$V_{\mathrm{soft}}(s_{t+k}) = \sum_a \pi_i[\min(Q_{\phi_1}, Q_{\phi_2}) - \alpha \log \pi_i] \tag{9}$$

where $\min(Q_{\phi_1}, Q_{\phi_2})$ uses the twin critics to reduce overestimation bias [3].

The parameter k controls the bias-variance tradeoff [4]: small k relies more on the critic (lower variance, higher bias); large k relies more on observed rewards (lower bias, higher variance). $G_t^{(k)}$ is computed entirely from past experience — no future information is used at execution time.

C. Temporal Counterfactual Advantage

We replace $Q(s_t, \mathbf{a}_t)$ in the COMA advantage (Eq. 2) with $G_t^{(k)}$:

$$C_i^t = G_t^{(k)} \;-\; B_i(s_t,\, \mathbf{a}_{-i}^t) \tag{10}$$

where B_i is the same COMA counterfactual baseline (Eq. 1), reusing the same centralized critic. C_i^t thus captures: *how much better (or worse) were actual long-term outcomes compared to what agent i would have been expected to contribute, given what everyone else did?*

Algorithm 1 TCCA-SAC Training

Initialize actors $\{\pi_i\}$, twin critics Q_{ϕ_1}, Q_{ϕ_2}, target critics $Q_{\bar{\phi}_1}, Q_{\bar{\phi}_2}$, replay buffer $\mathcal{D}$
for each episode **do**
  Reset hidden states $h_i \leftarrow \mathbf{0}$ for all agents
  for each timestep t **do**
    Each agent i samples $a_i^t \sim \pi_i(\cdot \mid o_i^t, h_i)$, updates h_i
    Execute joint action $\mathbf{a}^t$; observe R_t, s_{t+1}
    Store $(s_t, \mathbf{a}^t, R_t, s_{t+1}, \{h_i\}, \text{avail})$ in $\mathcal{D}$
  end for
  if $|\mathcal{D}| \geq$ warmup **then**
    for each training update **do**
      Sample k-step segment τ_t from $\mathcal{D}$
      Compute $G_t^{(k)}$ via Eq. (7) with soft bootstrap (Eq. 9)
      for each agent i **do**
        Compute baseline B_i via Eq. (1)
        Compute advantage $C_i^t \leftarrow G_t^{(k)} - B_i$
        Update π_i via $\nabla J(\pi_i)$ using Eq. (11)
      end for
      Update Q_{ϕ_1}, Q_{ϕ_2} via MSE$(G_t^{(k)})$
      Polyak update: $\bar{\phi} \leftarrow \tau\phi + (1-\tau)\bar{\phi}$
    end for
  end if
end for

D. Policy Update

Each agent's policy is updated using the SAC objective, substituting C_i^t as the advantage signal:

$$J(\pi_i) = \mathbb{E}\big[C_i^t \; - \; \alpha \log \pi_i(a_i^t \mid o_i^t)\big] \tag{11}$$

The entropy coefficient α is retained from SAC, ensuring stable exploration throughout training. The twin critics are trained via MSE loss against the k-step soft target $G_t^{(k)}$.

E. Off-Policy Correction

Since trajectory segments in the replay buffer were collected under earlier policy versions, $G_t^{(k)}$ is technically off-policy. We adopt two mitigations. First, the bootstrapped k-step return structure (Eq. 7) reduces the staleness window compared to full Monte Carlo returns. Second, we rely on SAC's inherent robustness to off-policy data as the primary correction mechanism; prioritized replay is treated as an optional hyperparameter and not used in the main experiments.

F. Full Algorithm

Algorithm 1 presents the complete TCCA-SAC training procedure.

IV. Experimental Setup

A. Environments

MPE simple_spread_v3. Three agents navigate a 2D continuous space to cover three target landmarks. Reward is shared and penalizes the minimum distance between any landmark and its nearest agent. This environment is fully observable, admits fast training, and serves as the primary sanity-check benchmark.

SMAC 3m. Three allied Marines must defeat three enemy Marines in the StarCraft II engine. Observations are partial (local field-of-view only), actions are discrete (move, attack target, stop, no-op), and some actions are unavailable (e.g., attacking an out-of-range unit). Episodes are sparse-reward: a small per-kill reward and a large bonus for winning. This environment tests whether TCCA scales to the recurrent, partially-observable, action-masked setting.

TABLE I: Hyperparameters for All Experimental Configurations

Parameter	COMA MPE	TCCA MPE	TCCA-SAC MPE	TCCA-SAC SMAC
Episodes	50k	50k	50k	50k
Batch size	32 ep	32 ep	64 seg	32 seg
γ	0.99	0.99	0.99	0.99
LR (actor)	3e-4	3e-4	3e-4	5e-4
LR (critic)	1e-3	1e-3	1e-3	5e-4
k (horizon)	1	**5**	**3**	**3**
α (entropy)	0.01	0.01	**0.005**	**0.05**
τ (Polyak)	—	—	0.005	0.005
Replay buffer	—	—	50k ep	5k ep
Critic type	Single	Single	Twin	Twin
Actor type	MLP	MLP	MLP	GRU

B. Algorithm Variants

We implement and compare five configurations:

- **COMA-MPE**: Vanilla COMA with TD(0), on-policy, independent MLP actors.
- **TCCA-MPE**: TCCA with k=5, on-policy, same MLP architecture as COMA-MPE.
- **TCCA-SAC-MPE**: TCCA-SAC with twin critics, replay buffer, entropy α=0.005, k=3.
- **COMA-SMAC**: COMA with GRU actors, Monte Carlo returns, action masking, on-policy.
- **TCCA-SAC-SMAC**: Full research build — GRU actors, twin critics, k-step soft returns, stored hidden states, validity masks, α=0.05, k=3.

C. Hyperparameters

Table I lists the hyperparameters for all configurations.

D. Network Architectures

MLP Actor. Two-layer MLP with hidden size 64, ReLU activations, softmax output over discrete actions.

GRU Actor (SMAC). Linear input projection $\rightarrow$ GRUCell (hidden=64) $\rightarrow$ linear output. Input is $[\text{obs} \,\|\, \text{last_action_onehot} \,\|\, \text{agent_id_onehot}]$. Unavailable actions are masked to -10^{10} before softmax.

Central Critic. MLP with two hidden layers of size 64. Input is $[\text{global_state} \,\|\, \text{other_agents_actions_onehot} \,\|\, \text{agent_id_onehot}]$, output is per-action Q-values for the evaluated agent.

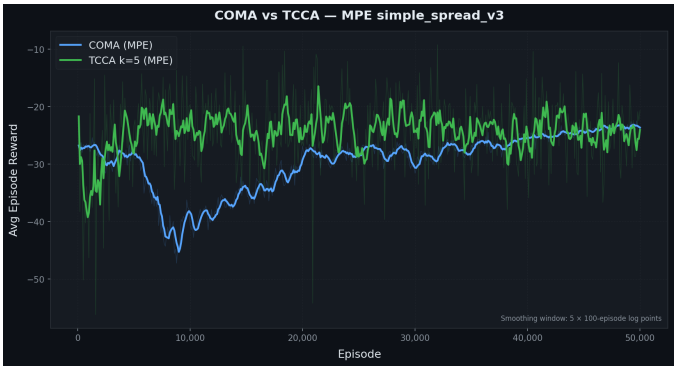


Fig. 1: COMA vs. TCCA (k=5) on MPE `simple_spread_v3`. TCCA converges stably to $\sim$−21 from episode $\sim$2,000 while COMA undergoes a catastrophic critic crash between episodes 7,000–12,000 before slowly recovering to $\sim$−23.

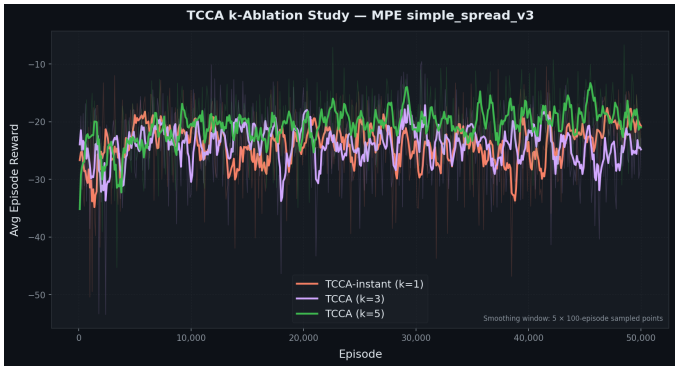


Fig. 2: k-ablation on MPE. Reward improves monotonically with horizon k, confirming that longer temporal returns provide a more informative credit signal. TCCA-instant (k=1) is equivalent to COMA with a replay buffer.

Twin Critics. Two independent central critics trained in parallel; $\min(Q_{\phi_1}, Q_{\phi_2})$ is used for bootstrapping and the counterfactual baseline.

E. Ablation Studies

Two ablations are conducted to isolate the contribution of each component:

Temporal horizon k. We train three instances of TCCA-MPE with $k \in \{1, 3, 5\}$. k=1 (TCCA-instant) is equivalent to COMA with an off-policy replay buffer, providing a clean lower bound on the temporal component.

Entropy regularization α. We train TCCA-SAC-MPE with α=0 (entropy disabled) and α=0.005 (standard), holding all other parameters constant.

V. Results

A. COMA vs. TCCA on MPE

Fig. 1 shows the 100-episode rolling average reward for COMA and TCCA (k=5) on MPE over 50,000 episodes. TCCA reaches near-optimal performance ($\sim$−21) from approximately episode 2,000 and maintains it stably throughout training. COMA, by contrast, suffers a severe critic instability between episodes 7,000 and 12,000, crashing to $\sim$−47 before slowly recovering. This crash is consistent with the known sensitivity of TD(0)-based critics: when early Q-value estimates are inaccurate, the actor updates in harmful directions, creating a self-reinforcing degradation cycle. By episode 50,000, COMA recovers to approximately −23, but never reaches TCCA's final performance level.

B. Temporal Horizon Ablation

Fig. 2 shows the reward curves for TCCA-instant (k=1), TCCA (k=3), and TCCA (k=5) on MPE. The ordering is monotonic: k=5 > k=3 > k=1 throughout training. Final 100-episode average rewards at episode 50,000 are approximately −21 (k=5), −23 (k=3), and −26 (k=1).

Notably, TCCA-instant (k=1) — which has a replay buffer but no multi-step return — already avoids COMA's severe crash, suggesting that off-policy data reuse provides some stabilization benefit independent of the temporal extension. However, the additional improvement from k=3 to k=5 demonstrates that the temporal return itself, not merely the replay buffer, is responsible for the performance gains.

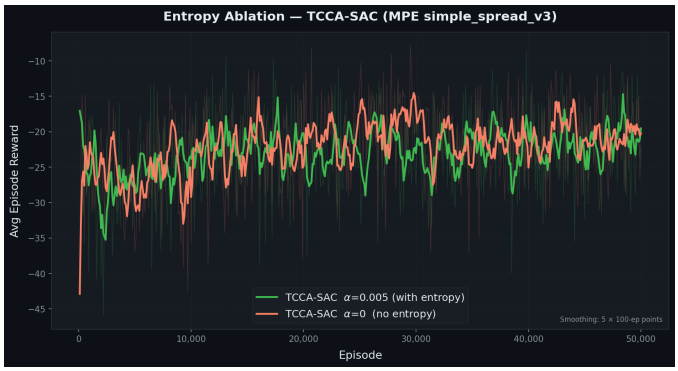


Fig. 3: Entropy ablation on MPE. Entropy regularization (α=0.005) reduces initial instability and accelerates early convergence. Final performance is similar, suggesting entropy is primarily a stabilizer on this task.

C. Entropy Regularization Ablation

Fig. 3 compares TCCA-SAC with α=0.005 versus α=0 on MPE. Both variants converge to similar final rewards ($\sim$−20 to −22), but the entropy-regularized run shows a shallower initial dip (peak crash depth −35 vs. −44 for α=0) and faster recovery in the first 5,000 episodes. This pattern is consistent with entropy regularization providing exploration insurance during the early phase of training, when the policy is highly uncertain and prone to premature convergence. On the simpler MPE task, where the action space is small, the long-run benefit of entropy is marginal — a finding consistent with the SAC literature on low-dimensional discrete action spaces.

D. COMA vs. TCCA-SAC on SMAC

Figs. 4 and 5 show win rate and average episode reward on SMAC `3m` over 50,000 episodes.

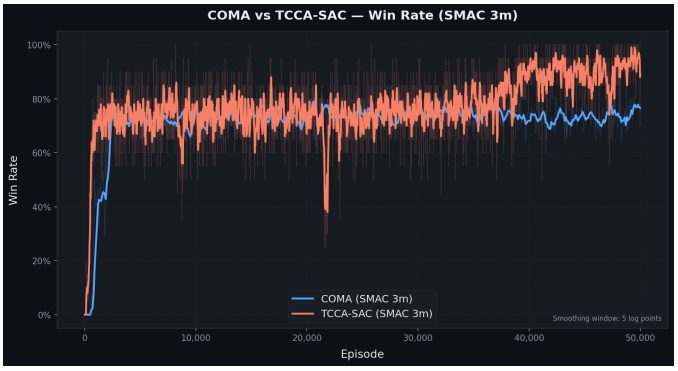


Fig. 4: Win rate on SMAC 3m. TCCA-SAC begins winning at episode ∼500 versus ∼1,500 for COMA, and converges to ∼90% by episode 40,000, compared to COMA's plateau of ∼75%.

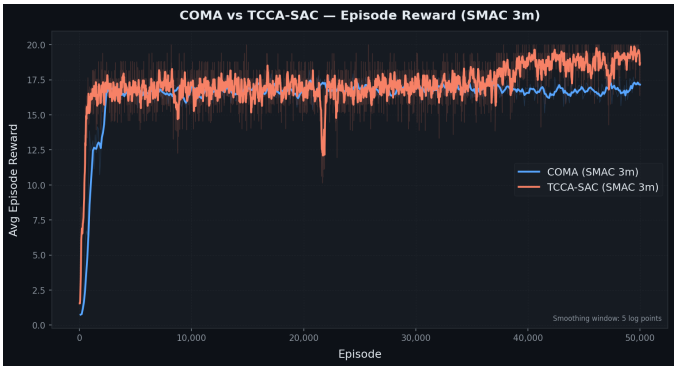


Fig. 5: Average episode reward on SMAC 3m. TCCA-SAC achieves higher reward throughout training and continues improving after episode 40,000 while COMA plateaus.

TCCA-SAC achieves a substantially faster initial rise: win rate reaches 60% by episode ∼500, whereas COMA requires ∼1,500 episodes to reach the same level. In the steady state (episodes 10,000–35,000), both methods operate in a similar range (∼72–80% win rate), but from episode 37,000 onward TCCA-SAC begins a second improvement phase, ultimately converging to ∼90–95% by episode 50,000 while COMA plateaus at ∼75%.

A transient policy collapse at episode ∼22,000 is visible in TCCA-SAC (win rate drops to ∼38% for ∼300 episodes), consistent with known instabilities that arise when the replay buffer undergoes a distribution shift from earlier policy checkpoints. The policy recovers within approximately 3,000 episodes, after which the improvement trend resumes.

The reward gap is more pronounced: TCCA-SAC averages ∼18–19 reward versus COMA's ∼17, and continues trending upward in the final 10,000 episodes while COMA remains flat.

E. Summary of Results

Table II summarizes the final performance metrics.

TABLE II: Final Performance at Episode 50,000

Method	Env	Avg Reward	Win Rate
COMA	MPE	-23	—
TCCA-instant (k=1)	MPE	-26	—
TCCA (k=3)	MPE	-23	—
TCCA (k=5)	MPE	-21	—
COMA	SMAC	∼17	∼75%
TCCA-SAC	SMAC	∼19	∼90%

VI. Discussion

A. Why the Temporal Return Helps

The policy gradient theorem [8] establishes that the optimal baseline to subtract from the gradient is the state-value function $V(s)$. COMA approximates this per-agent via the critic Q_ϕ. The key vulnerability is that Q_ϕ is inaccurate during early training, making the baseline actively harmful in the worst case — as evidenced by the crash in Fig. 1.

TCCA's k-step return reduces this vulnerability by anchoring the first k steps of the return to observed rewards, which are always accurate. The critic is only needed to estimate the value at the terminal step s_{t+k}, where it has been updated on k-step segments that are empirically more informative. The ablation in Fig. 2 confirms this: as k increases, the reliance on an accurate early critic decreases, and performance improves monotonically.

B. Role of SAC Components on SMAC

On SMAC, several components of the TCCA-SAC system interact:

- **GRU actors**: Essential for partial observability — without hidden state, agents cannot infer the full game state from local observations alone.
- **Twin critics**: Reduce Q-value overestimation, which is particularly damaging in sparse-reward combat tasks where inflated Q-values can cause aggressive premature termination of coordination.
- **Entropy regularization** (α=0.05): More important on SMAC than MPE due to the larger and more structured discrete action space (attack, move, stop, no-op × directions). Entropy prevents the policy from collapsing prematurely onto suboptimal attack patterns.
- **Stored hidden states in replay buffer**: Enables correct off-policy re-play of any trajectory segment without re-running the full episode, which would be computationally prohibitive.

C. Limitations and Future Work

The primary limitation is the absence of multiple independent seeds and confidence intervals on all reported curves. This is a standard requirement for rigorous MARL evaluation [9], and variability across seeds on SMAC can be substantial. We ran single seeds due to compute constraints; future work should report 5-seed averages with standard deviations.

A second limitation is the off-policy bias in $G_t^{(k)}$. While SAC provides robustness to moderate policy lag, importance

sampling corrections [10] or V-trace [11] could more formally address this. We treat the off-policy correction as a known limitation and an avenue for future work.

Extending TCCA to larger SMAC scenarios (`2s3z`, `5m_vs_6m`) and comparing against QMIX remain natural next steps.

VII. Conclusion

We have proposed and evaluated **Temporal Counterfactual Credit Assignment (TCCA)**, a principled extension to COMA that incorporates k-step discounted returns into the per-agent counterfactual advantage signal. By computing $G_t^{(k)}$ retrospectively from replay buffer trajectories, TCCA provides each agent with a credit signal that reflects long-term consequences of its actions, while avoiding any use of future information at execution time.

Integrated with Soft Actor-Critic's entropy regularization, twin critics, and Polyak target updates, TCCA-SAC provides a stable, sample-efficient framework for cooperative MARL. Empirically, TCCA avoids the critic crash that afflicts COMA on MPE and achieves superior sample efficiency and final win rate on SMAC `3m`. The k-ablation confirms that the performance gain is attributable to the temporal return extension, not merely to the replay buffer or architectural changes.

The core contribution is both conceptual and empirical: identifying the temporal gap in COMA's credit signal, formalizing the TCCA fix, and demonstrating its benefit across two cooperative benchmarks with controlled ablations.

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